

SEQUENCER

1

E \$\$ <3
Upload raw
vehicle CAN
bus logs
green

E \$\$\$ <3<3<3
View current
state-of-health
(SOH) percentage
green

E \$\$ <3<3
View calculated
remaining useful
life (RUL) hours
yellow

2

E \$\$ <3
Filter active fleet
by remaining
useful life
green

E \$\$ <3<3
Export high-
fidelity data
profile
summaries
yellow

MVP

EEE \$\$\$ <3<3<3
Map custom
telemetry channels
to AI model inputs
red

3

EE \$\$\$ <3<3<3
Set critical
threshold
alert triggers
green

EEE \$\$\$ <3<3<3
Connect cloud
database
stream
red

INCREMENT

4

EE \$\$\$ <3<3
View individual
truck accelerated
wear history
yellow

EEE \$\$\$ <3<3<3
Compare
software baseline
runs side-by-side
red

5

EEE \$\$\$ <3<3<3
View degradation
stress breakdown
chart
yellow

EE \$\$\$ <3<3<3
Overlay SOH
predictions vs
physical
potentization curves
red

☐ **SCRUM Sprint 1** 22 Jun – 6 Jul (5 work items) 24 0 0 Complete sprint 

Establish the foundational end-to-end MVP tool by enabling engineers to manually upload raw CAN bus logs and instantly view calculated fuel cell SOH% and RUL hours on a core analytics dashboard.

☒ SCRUM-1	Story 1.1: Manual Log File Upload Interface	EPIC 1: TELEMETRY D...	TO DO ▾	Jul 10	2	⬇	
☒ SCRUM-3	Story 1.2: CAN Bus Log Parsing & Column Mapping	EPIC 1: TELEMETRY D...	TO DO ▾	Jul 10	6	⬇	
☒ SCRUM-4	Story 2.1: Execute SOH and RUL Analytical Pipeline	EPIC 2: DURABILITY ...	TO DO ▾	Jul 10	7	⬇	
☒ SCRUM-7	Story 3.1: MVP Core Metrics View (SOH & RUL Dashboard)	EPIC 3: ENGINEERIN...	TO DO ▾	Jul 10	6	⬇	
☒ SCRUM-10	Story 4.2: High-Fidelity Engineering Profile Export	EPIC 4: FLEET ANALY...	TO DO ▾	Jul 17	3	=	

 Create

☐ **Backlog** (7 work items) 41 0 0  Create sprint

☒ SCRUM-9	Story 4.1: Fleet-Wide Filtering by Remaining Useful Life (RUL)	EPIC 4: FLEET ANALY...	TO DO ▾	Jun 26	7	=	
☒ SCRUM-12	Story 5.1: Dynamic Telemetry Channel Configuration Layer	EPIC 5: DYNAMIC IN...	TO DO ▾		6	=	
☒ SCRUM-13	Story 5.2: Live Cloud Database Stream Integration	EPIC 5: DYNAMIC IN...	TO DO ▾		8	=	
☒ SCRUM-15	Story 6.1: Configurable Critical Degradation Threshold Alerts	EPIC 6: AUTOMATED ...	TO DO ▾		3	=	
☒ SCRUM-17	Story 7.1: Individual Vehicle Accelerated Wear History Timeline	EPIC 7: ADVANCED D...	TO DO ▾		5	=	
☒ SCRUM-18	Story 7.3: Degradation Stress Component Breakdown Visualization	EPIC 7: ADVANCED D...	TO DO ▾		4	=	
☒ SCRUM-19	Story 7.4: Virtual SOH Predictions vs. Physical Polarization Curves Overlay	EPIC 7: ADVANCED D...	TO DO ▾		8	=	

Fuel Cell Durability Estimator ...

Summary Backlog Board Code **Timeline** Docs Forms Development +

GL
Person icon
Epic ▾
Label ▾
Status category ▾

Work	May	June	July
Sprints			
☐ > SCRUM-2 EPIC 1: Telemetry Data Ingestion Pipeline		SCRUM Sprint 1	
☐ > SCRUM-5 EPIC 2: Durability Calculation Engine Integration			
☐ > SCRUM-6 EPIC 3: Engineering Analytics Dashboard			
☐ > SCRUM-8 EPIC 4: Fleet Analytics & Reporting			
☐ > SCRUM-11 EPIC 5: Dynamic Ingestion & Cloud Data Streaming			
☐ > SCRUM-14 EPIC 6: Automated Health Monitoring & Alerting			
☐ > SCRUM-16 EPIC 7: Advanced Durability Diagnostics & Validation			
+ Create Epic			

Story 1.1: Manual Log File Upload Interface



Description

- **Summary:** As Carlos, I want to manually upload static CAN bus log files via a web interface so that the platform can queue the data for post-processing.
- **Description:** The system needs a simple, foolproof drop-zone or file-picker component that accepts localized engineering log formats.
- **Acceptance Criteria:**
 - **Given** Carlos is on the platform upload screen, **when** he drags and drops a raw `.csv` or `.blf` or `.mdf` or `mf4` log file, **then** the system validates the file format and size limits.
 - **Given** a valid file is uploaded, **when** the upload completes, **then** the system displays a success state and passes the file path to the processing queue.
 - **Given** an invalid file extension or corrupted file is uploaded, **when** evaluated, **then** the UI displays an explicit engineering error message detailing the failure.

Subtasks

Add subtask

Linked work items

blocks



To Do



Improve Task

Your pinned fields

Definition of Ready

Maximum file size limit (e.g., 500MB) and allowed extensions are explicitly specified.

- UI/UX wireframe or layout for the drag-and-drop zone is approved.
- Target storage repository (sandbox cloud bucket or local directory) is provisioned and accessible.
- Story is estimated in story points by the development team.

Definition of Done (DoD)

- Upload component successfully accepts, validates, and stores a sample engineering log.
- Error handling code successfully catches and displays messages for corrupted or oversized files.
- Code passes peer review, satisfies linting standards, and is merged into the development branch.
- UI component is verified to render properly across standard engineering browsers.

Story 1.2: CAN Bus Log Parsing & Column Mapping



Description

- Summary:** As a System Engine, I want to parse raw CAN bus signals from the uploaded log file so that standard telemetry channels are mapped accurately to the backend data structures.
- Acceptance Criteria:**
 - Given** an uploaded CSV/MDF/MF4/BLF log file, **when** parsed by the data layer, **then** it extracts core parameters (e.g., stack current, voltage, cell temperatures, operating hours, load use profile, etc).
 - Given** missing or misaligned column headers in the file, **when** parsed, **then** the system falls back to default index mapping or logs a parsing exception.

Subtasks

Add subtask

Linked work items

blocks

☒ SCRUM-4 Story 2.1: Execute SOH and RUL Analytical Pipeline

TO DO

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is blocked by

☒ SCRUM-1 Story 1.1: Manual Log File Upload Interface

TO DO

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To Do

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✦ Improve Task

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Definition of Ready • Unsaved changes

- The explicit data schema or target signal mapping spreadsheet is finalized (defining precise header names for current, voltage, temperature, and hours).
- At least three real-world sample log files from recent vehicle testing are available for parsing tests.
- The target database or data structure configuration is finalized.

Definition of Done (DoD)

- The parser successfully extracts all mandatory columns from the sample log files without throwing exceptions.
- Unit tests cover edge cases, such as handling null values, missing timestamps, or unexpected data types.
- Data parsing output has been peer-reviewed and verified by the data scientist to ensure data integrity matches downstream requirements.

Details



Assignee

GL Guilherme Alves Lima

Story 2.1: Execute SOH and RUL Analytical Pipeline



Description

- Summary:** As a Fuel Cell Stack Scientist, I want to execute the fuel cell durability algorithm against the parsed telemetry table so that the tool calculates current State-of-Health (SOH) and Remaining Useful Life (RUL).
- Description:** Run the post-processing script to evaluate stack wear parameters against baseline degradation metrics.
- Acceptance Criteria:**
 - Given** a successfully parsed CAN bus dataset, **when** the durability calculation engine completes execution, **then** it generates a scalar value for SOH (percentage) and RUL (remaining operational hours).
 - Given** the calculation completes, **when** generating outputs, **then** the processing runtime must execute in under 120 seconds for typical engineering log lengths.

Subtasks

Add subtask

Linked work items

is blocked by

☒ SCRUM-3 Story 1.2: CAN Bus Log Parsing & Column Mapping

TO DO

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To Do

Improve Task

Your pinned fields

Definition of Ready

- The core fuel cell degradation algorithm script is finalized, packaged, and verified locally by Data Science using standalone mock data.
- The exact mathematical inputs and required boundary limits for the algorithm are documented.
- Story 1.2 (Parsing) is completed or mature enough to provide test data frames.

Definition of Done (DoD)

- The calculation pipeline executes end-to-end, converting the parsed data into an accurate SOH percentage and RUL hourly figure.
- Processing performance test confirms execution time is under 120 seconds for a standard log file.
- Calculated outputs are validated against verified physical laboratory baseline data to ensure they sit within the expected error margin (e.g. 5%).

Story 3.1: MVP Core Metrics View (SOH & RUL Dashboard)



Description • Unsaved changes

- **Summary:** As Carlos, I want to view a high-level summary dashboard displaying the calculated SOH% and RUL hours so that I can instantly evaluate the durability impact of my calibration changes.
- **Description:** A clean, minimal UI dedicated entirely to exposing the two primary performance indicators calculated by the engine.
- **Acceptance Criteria:**
 - **Given** the calculation engine has completed processing an uploaded log file, **when** Carlos opens the results view, **then** the interface renders the current State-of-Health clearly as a percentage figure.
 - **Given** the dashboard view is rendered, **when** displaying the stack lifetime remaining, **then** the Remaining Useful Life (RUL) must be explicitly presented as total operational hours.

Subtasks

Add subtask

Linked work items

Add linked work item

Activity



To Do

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▼ Your pinned fields

Definition of Ready

- Dashboard visual layout (metric cards, font sizes, and arrangement) is reviewed and approved by the UX designer and product owner.
- The calculation outputs from Story 2.1 are accessible via a defined data object or lightweight database query.
- Frontend tech stack or dashboard framework is configured and ready for component deployment.

Definition of Done (DoD)

- The web dashboard correctly pulls and displays the dynamic SOH % and RUL hours upon file processing completion.
- An end-to-end integration test is successfully performed by the team: Uploading a file → Processing → Viewing correct dashboard results.
- The UI layout is officially reviewed and accepted by the Product Owner against the initial MVP canvas layout requirements.

Story 4.2: High-Fidelity Engineering Profile Export



▼ Description • Unsaved changes

- **Summary:** As Sarah, I want to export structured data summaries (CSV/Parquet) of processed degradation runs so that I can feed the results into downstream validation toolchains.
- **Acceptance Criteria:**
 - **Given** a completed log analysis view, **when** Sarah clicks "Export Profile Summary", **then** the system generates a downloadable file containing calculated SOH and rolling degradation vectors.

Subtasks

Add subtask

Linked work items

Add linked work item

Activity

- All Comments History Work log



Add a comment...

- Status update... Thanks... Agree...

To Do </> ⚡ ⚙ Improve Task

▼ Your pinned fields

Definition of Ready
Schema definition for the export payload is finalized.

Definition of Done (DoD)
File exports successfully without corruption and maintains data type integrity

▼ Details ⚙

Assignee	Guilherme Alves Lima
Priority	🔴 Medium
Parent	🔗 SCRUM-8 EPIC 4: Fleet Analytics & Rep
Due date	📅 Jul 17, 2026
Labels	Analytics Reporting
Team	Fuel Cell Software Team